

# Skills Summary

## Comparing Models, Qualifying Conclusions

*Use this reference while completing your worksheet or when studying.*

### Skill 1 — Choosing a Summary: Mean or Median

**Two models for “typical”.** The mean adds every value and divides, so one extreme value moves it a long way. The median is the middle value in order, so an extreme value moves it hardly at all. Report the mean with the standard deviation; report the median with the interquartile range. When the two summaries disagree by a lot, an outlier is usually the reason — say so rather than picking silently.

**Example:** Five days of rainfall: 3, 5, 6, 8, 28 mm. Compute both summaries and compare.

**Step 1.** The list has one value far above the rest, so the mean and the median will not agree — compute both before choosing.

**Step 2.** Mean =  $\frac{3+5+6+8+28}{5} = \frac{50}{5} = 10$ ; median is the third value in order, 6. The 28 mm day is what pulls the mean up.

⇒ mean **10**, median **6** — report the median for a typical day

**Try it:** Five days: 1, 3, 4, 6, 26 mm. Compute the mean and the median.

check: mean 8, median 4

### Skill 2 — Comparing Fits With Residuals

**Rule:** for each data point, residual =  $y - \hat{y}$ , the observed value minus the model’s prediction. A negative residual means the model predicted too high. Square every residual and add: that single number scores the model, and between two models the *smaller* score is the closer fit. Squaring is what stops over- and under-predictions cancelling out.

**Example:** Model  $\hat{y} = 2 + 2x$  against the points (1, 5), (2, 7), (3, 7).

**Step 1.** Scoring a fit means comparing each observation with what the model predicts there, so compute all three predictions first.

**Step 2.** Predictions 4, 6, 8 give residuals  $5 - 4 = 1$ ,  $7 - 6 = 1$ ,  $7 - 8 = -1$ ; squaring and adding gives  $1 + 1 + 1 = 3$ .

⇒ first residual **1**, fit score **3**

**Try it:** Model  $\hat{y} = 1 + 3x$  against (1, 6), (2, 8), (3, 9). Find the residual at  $x = 1$  and the fit score.

check: residual 2, fit score 9

### Skill 3 — Saying Only What the Data Supports

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**Three limits to state on every conclusion.** **Who** — a proportion describes the group that was actually surveyed, and reaches further only if that group was randomly sampled from the wider one. **Why not** — observational data show that two things go together, never that one causes the other; name a confounding variable that could explain both. **How far** — a fitted model is trustworthy inside the range of the data; predicting outside it is extrapolation, and two models that agree inside can disagree badly outside.

**Example:** Of 60 surveyed members of a school's chess club, 24 walk to school. What proportion is that, and who does it describe?

**Step 1.** The count is a proportion of the group actually asked, so compute it and then name that group before saying anything wider.

**Step 2.**  $\frac{24}{60} = \frac{2}{5}$ . It describes the chess club members surveyed — club members are not a random sample of the school, so it says nothing about the school as a whole.

$\Rightarrow$   $\frac{2}{5}$  of the surveyed club members

**Try it:** Of 90 students leaving the library, 27 took the bus home. What proportion is that?

check:  $\frac{27}{90}$  of the students surveyed